LÝ THUYẾT MIN3

PHẦN TỬ MỚI CS-MIN3 VÀ ES-MIN3

Trình bày: Phùng Văn Phúc

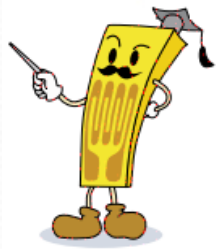
GVHD: TS. Nguyễn Thời Trung

Tp. HCM 18-08-2011

Nội dung trình bày

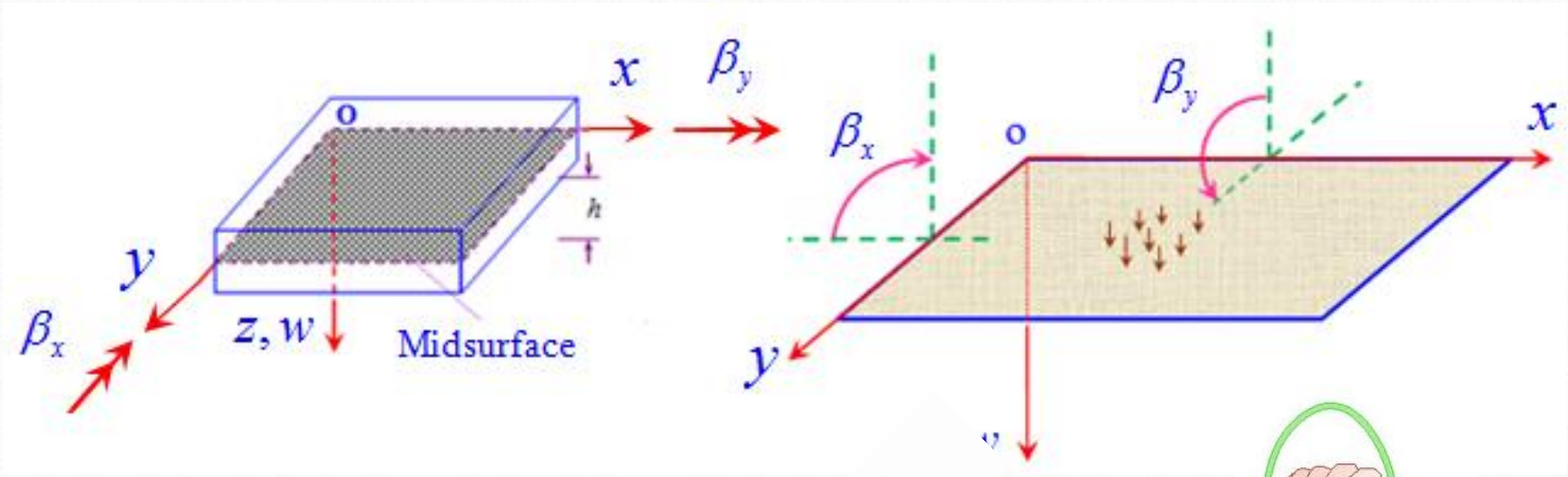


- Quy ước dấu trong tấm MIN3
- Thành lập các ma trận B
- Phần tử CS-MIN3
- Phần tử ES-MIN3
- Một số kết quả số



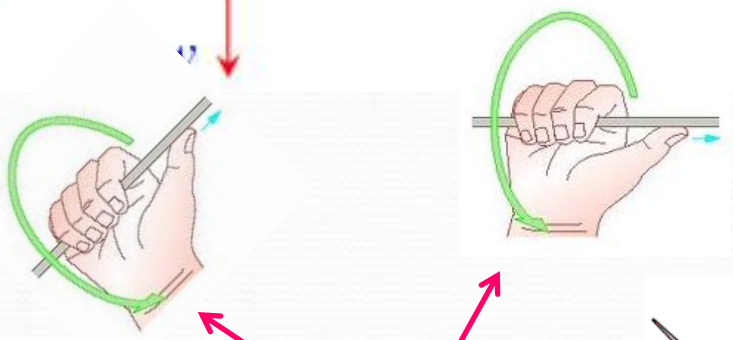
Phần tử MIN3

➢ Quy ước dấu trong tấm MIN



- β_y là góc quay xung quanh trục x
- β_x là góc quay xung quanh trục y

➔ Xác định theo quy tắc bàn tay phải



Chiều quay của β theo độ võng

Phần tử MIN3

➢ Biến xấp xỉ

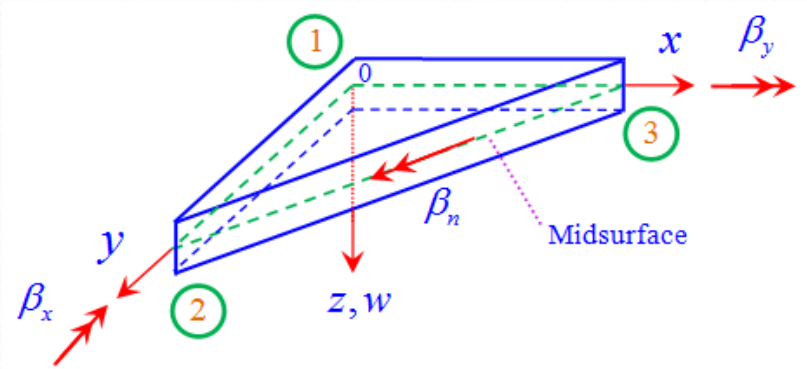
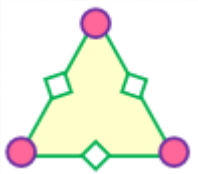
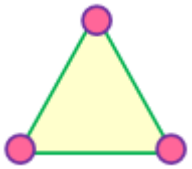


Table 1: Nodal configurations for initial (unconstrained) and constrainted displacement field in the MIN3

Shape functions		Initial nodal configuration	Continuous shear edge constraints: $\left(w_{,s}+\beta_n\right)_{,s}\Big\|_{\text{edges}}=0$	Constrained nodal configuration
w	β_x, β_y			
Quadratic	Linear		Three edge constraints ⇨	

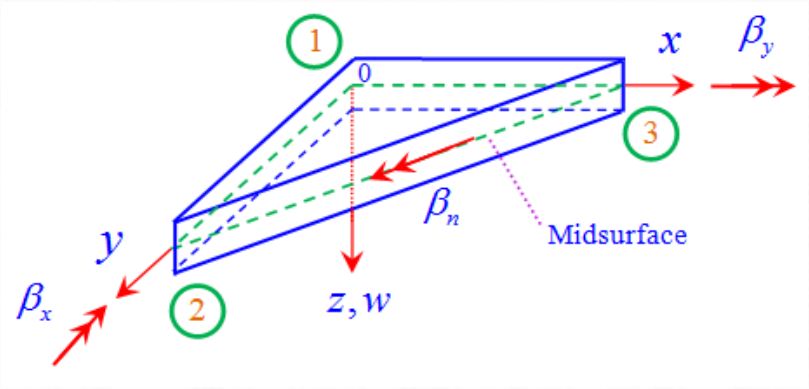
Phần tử MIN3



➢ Biến xấp xỉ

○ Góc xoay

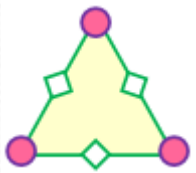
$$\beta_x = N_i \beta_{xi} \qquad \beta_y = N_i \beta_{yi} \qquad i = 1,2,3$$



➔ Giống DSG3, N_i hàm dạng 3 node

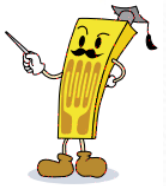
○ Độ võng

$$w = \sum_{I=1}^{6} R_I w_I = \mathbf{R}\mathbf{w}_{\text{ini}}$$



$$\mathbf{R} = \begin{bmatrix} N_1(2N_1-1) & N_2(2N_2-1) & N_3(2N_3-1) & 4N_1N_2 & 4N_2N_3 & 4N_3N_1 \end{bmatrix}$$

$$\mathbf{w}_{\text{ini}}^T = \begin{bmatrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 \end{bmatrix}$$



Phần tử MIN3



- Điều kiện ràn buộc của 3 node giữa

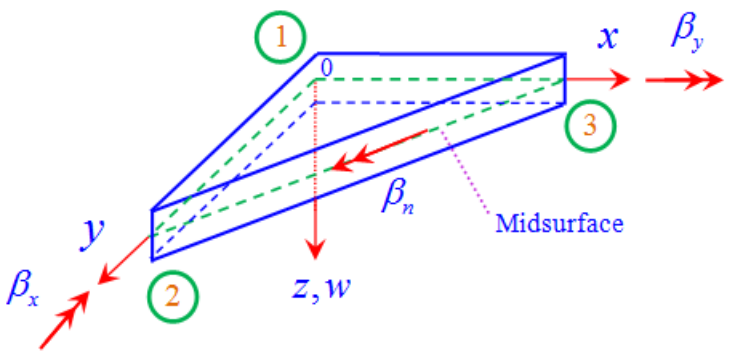
$$\left(w_{,s}+\beta_n\right)_{,s}\Big|_{\text{edges}}=0$$

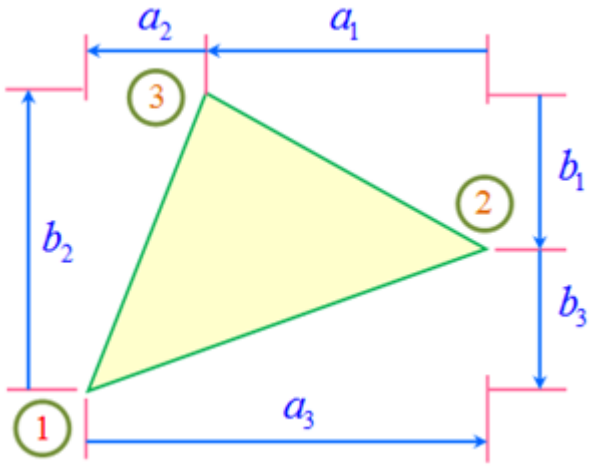
$$w_4=\frac{1}{2}\left(w_1+w_2\right)+\frac{1}{8}\left[b_3\left(\beta_{x1}-\beta_{x2}\right)+a_3\left(\beta_{y2}-\beta_{y1}\right)\right]$$

$$w_5=\frac{1}{2}\left(w_2+w_3\right)+\frac{1}{8}\left[b_1\left(\beta_{x2}-\beta_{x3}\right)+a_1\left(\beta_{y3}-\beta_{y2}\right)\right]$$

$$w_6=\frac{1}{2}\left(w_3+w_1\right)+\frac{1}{8}\left[b_2\left(\beta_{x3}-\beta_{x1}\right)+a_2\left(\beta_{y1}-\beta_{y3}\right)\right]$$

$$w=\sum_{I=1}^{3}N_I w_I+\sum_{I=1}^{3}H_I\beta_{xI}+\sum_{I=1}^{3}L_I\beta_{yI}=\mathbf{Nw}+\mathbf{H}\boldsymbol{\beta}_x+\mathbf{L}\boldsymbol{\beta}_y$$





$$\mathbf{w}^T=\begin{bmatrix}w_1 & w_2 & w_3\end{bmatrix}\quad \mathbf{H}=\begin{bmatrix}H_1 & H_2 & H_3\end{bmatrix}\quad \mathbf{L}=\begin{bmatrix}L_1 & L_2 & L_3\end{bmatrix}$$

$$H_1=\frac{1}{2}\left(a_2N_3N_1-a_3N_1N_2\right)\quad H_2=\frac{1}{2}\left(a_3N_1N_2-a_1N_2N_3\right)\quad H_3=\frac{1}{2}\left(a_1N_2N_3-a_2N_3N_1\right)$$

$$L_1=\frac{1}{2}\left(b_3N_1N_2-b_2N_3N_1\right)\quad L_2=\frac{1}{2}\left(b_1N_2N_3-b_3N_1N_2\right)\quad L_3=\frac{1}{2}\left(b_2N_3N_1-b_1N_2N_3\right)$$



Phần tử MIN3

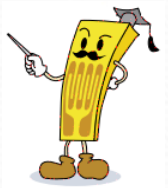


- Ma trận độ cứng

$$\mathbf{K}_e^{\text{MIN3}} = \int_{\Omega_e} \mathbf{B}^T \mathbf{D}^b \mathbf{B} \mathrm{d}\Omega + \int_{\Omega_e} \mathbf{S}^T \mathbf{D}^s \mathbf{S} \mathrm{d}\Omega$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & \mathbf{N}_{,x} \\ 0 & \mathbf{N}_{,y} & 0 \\ 0 & \mathbf{N}_{,x} & \mathbf{N}_{,y} \end{bmatrix} \qquad \mathbf{S} = \begin{bmatrix} \mathbf{N}_{,x} & \mathbf{H}_{,x} + \mathbf{N} & \mathbf{L}_{,x} \\ \mathbf{N}_{,y} & \mathbf{H}_{,y} & \mathbf{L}_{,y} + \mathbf{N} \end{bmatrix}$$

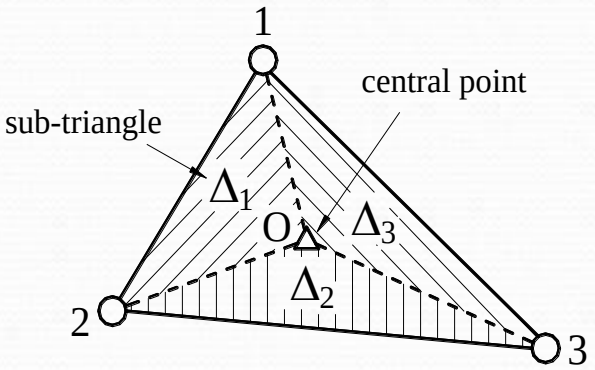
$$\hat{\mathbf{D}}^s = \frac{kt^3}{t^2 + \alpha h_e^2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \mathbf{D}^b = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$



Phần tử CS-MIN3

- Ma trận độ cứng
  - Tại tâm O

$$\mathbf{d}_{eO} = \frac{1}{3}\left(\mathbf{d}_{e1} + \mathbf{d}_{e2} + \mathbf{d}_{e3}\right)$$



  - Trong từng tam giác con sub-triangle

$$\boldsymbol{\kappa}^{\Delta_1} = \underbrace{\begin{bmatrix}\mathbf{b}_1^{\Delta_1} & \mathbf{b}_2^{\Delta_1} & \mathbf{b}_3^{\Delta_1}\end{bmatrix}}_{\mathbf{b}^{\Delta_1}} \begin{bmatrix}\mathbf{d}_{eO}\\ \mathbf{d}_{e1}\\ \mathbf{d}_{e2}\end{bmatrix} = \mathbf{b}^{\Delta_1}\mathbf{d}^{\Delta_1}$$

$$\boldsymbol{\kappa}^{\Delta_1} = \underbrace{\begin{bmatrix}\frac{1}{3}\mathbf{b}_1^{\Delta_1} + \mathbf{b}_2^{\Delta_1} & \frac{1}{3}\mathbf{b}_1^{\Delta_1} + \mathbf{b}_3^{\Delta_1} & \frac{1}{3}\mathbf{b}_1^{\Delta_1}\end{bmatrix}}_{\mathbf{B}^{\Delta_1}} \begin{bmatrix}\mathbf{d}_{e1}\\ \mathbf{d}_{e2}\\ \mathbf{d}_{e3}\end{bmatrix} = \mathbf{B}^{\Delta_1}\mathbf{d}_e$$

$$\boldsymbol{\gamma}^{\Delta_1} = \underbrace{\begin{bmatrix}\mathbf{s}_1^{\Delta_1} & \mathbf{s}_2^{\Delta_1} & \mathbf{s}_3^{\Delta_1}\end{bmatrix}}_{\mathbf{s}^{\Delta_1}} \begin{bmatrix}\mathbf{d}_{eO}\\ \mathbf{d}_{e1}\\ \mathbf{d}_{e2}\end{bmatrix} = \mathbf{s}^{\Delta_1}\mathbf{d}^{\Delta_1}$$

$$\boldsymbol{\gamma}^{\Delta_1} = \underbrace{\begin{bmatrix}\frac{1}{3}\mathbf{s}_1^{\Delta_1} + \mathbf{s}_2^{\Delta_1} & \frac{1}{3}\mathbf{s}_1^{\Delta_1} + \mathbf{s}_3^{\Delta_1} & \frac{1}{3}\mathbf{s}_1^{\Delta_1}\end{bmatrix}}_{\mathbf{S}^{\Delta_1}} \begin{bmatrix}\mathbf{d}_{e1}\\ \mathbf{d}_{e2}\\ \mathbf{d}_{e3}\end{bmatrix} = \mathbf{S}^{\Delta_1}\mathbf{d}_e$$

  - Trơn hóa CS

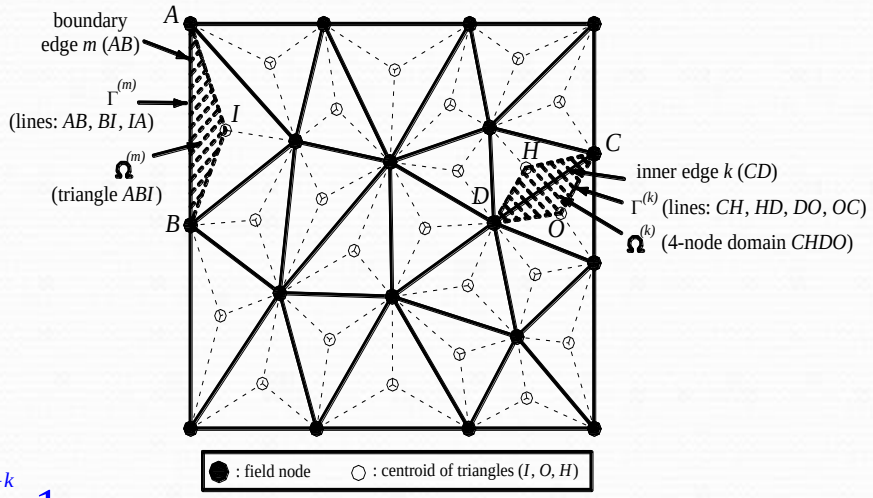
$$\tilde{\boldsymbol{\kappa}}_e = \frac{A_{\Delta_1}\boldsymbol{\kappa}^{\Delta_1} + A_{\Delta_2}\boldsymbol{\kappa}^{\Delta_2} + A_{\Delta_3}\boldsymbol{\kappa}^{\Delta_3}}{A_e}$$

$$\tilde{\boldsymbol{\gamma}}_e = \frac{A_{\Delta_1}\boldsymbol{\gamma}^{\Delta_1} + A_{\Delta_2}\boldsymbol{\gamma}^{\Delta_2} + A_{\Delta_3}\boldsymbol{\gamma}^{\Delta_3}}{A_e}$$

Phần tử ES-MIN3

- Ma trận độ cứng
  - Tại mỗi cạnh

$$\tilde{\boldsymbol{\kappa}}_k = \sum_{I=1}^{N_n^k} \tilde{\mathbf{B}}_I^b(\mathbf{x}_k)\mathbf{d}_I \qquad \tilde{\boldsymbol{\gamma}}_k = \sum_{I=1}^{N_n^k} \tilde{\mathbf{B}}_I^s(\mathbf{x}_k)\mathbf{d}_I$$



  - Trong miền trơn

$$\tilde{\mathbf{B}}_I^b(\mathbf{x}_k) = \frac{1}{A^{(k)}} \sum_{i=1}^{N_e^k} \frac{1}{3} A_i \mathbf{B}_i^b \qquad \tilde{\mathbf{B}}_I^s(\mathbf{x}_k) = \frac{1}{A^{(k)}} \sum_{i=1}^{N_e^k} \frac{1}{3} A_i \mathbf{B}_i^s$$

  - Độ cứng trơn hóa

$$\tilde{\mathbf{K}}^{(k)} = \int_{\Omega^{(k)}} (\tilde{\mathbf{B}}^b)^T \mathbf{D}^b \tilde{\mathbf{B}}^b \mathrm{d}\Omega + \int_{\Omega^{(k)}} (\tilde{\mathbf{B}}^s)^T \hat{\mathbf{D}}^s \tilde{\mathbf{B}}^s \mathrm{d}\Omega$$

$$= (\tilde{\mathbf{B}}^b)^T \mathbf{D}^b \tilde{\mathbf{B}}^b A^{(k)} + (\tilde{\mathbf{B}}^s)^T \hat{\mathbf{D}}^s \tilde{\mathbf{B}}^s A^{(k)}$$



LÝ THUYẾT MIN3

MỘT SỐ KẾT QUẢ ES-MIN3



Kết quả số

➢ Patch test

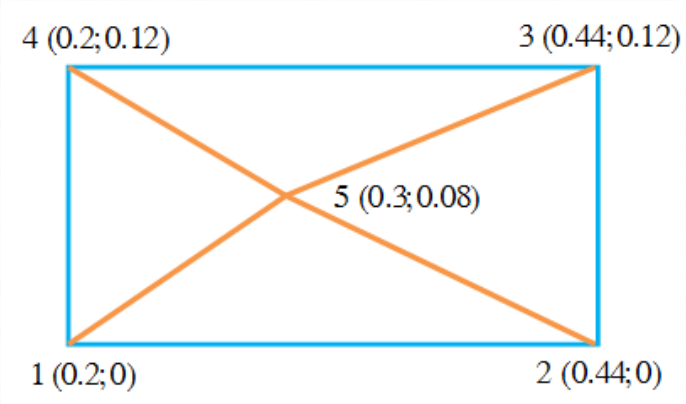
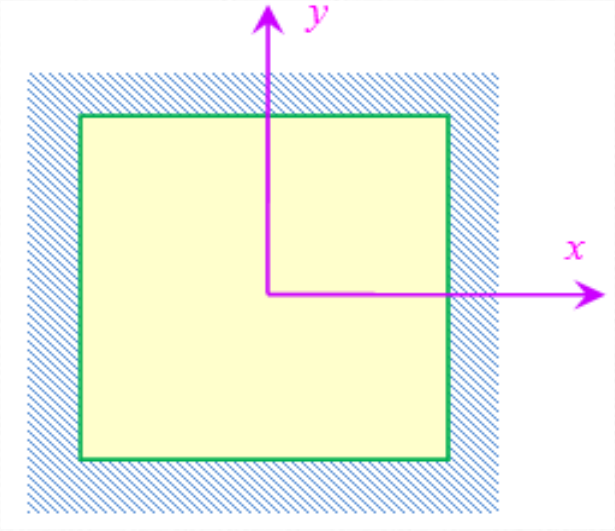
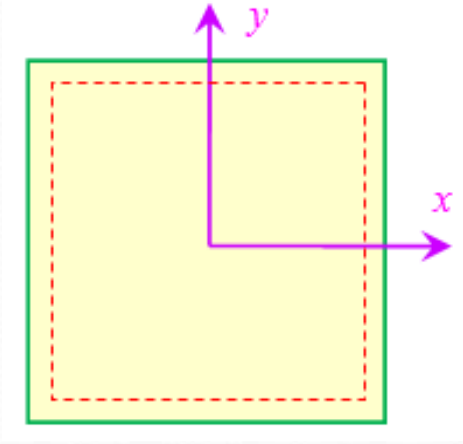


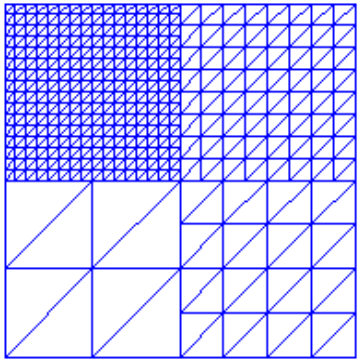
Table 1: Patch test.

Element	w_5	θ_{x5}	θ_{y5}	m_{x5}	m_{y5}	m_{xy5}
MIN3	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
DSG3	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
ES-DSG3	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
MITC4	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
ES-MIN3	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033
Exact	0.6422	1.1300	-0.6400	-0.0111	-0.0111	-0.0033

Tấm vuông

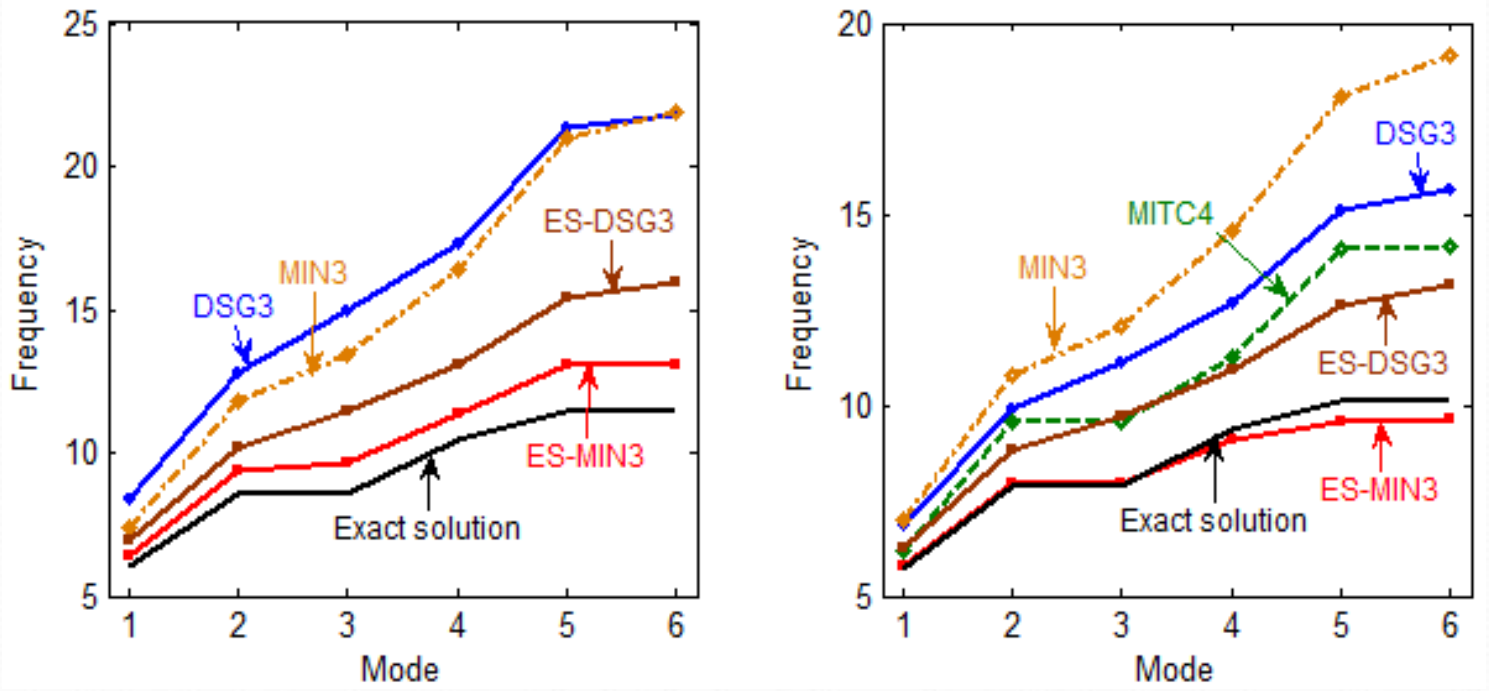






- ✓ Phân tích dynamic (mode)
- ✓ Phân tích static

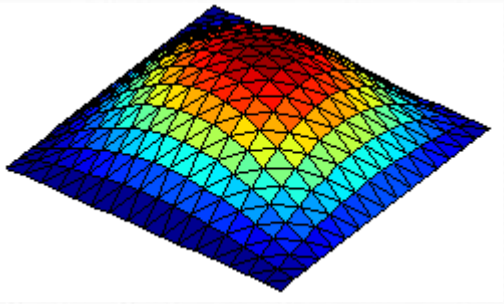
Phân tích mode tấm vuông



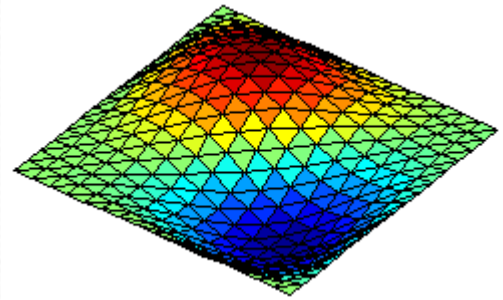
(a) (b)

Six lowest frequencies of CCCC square plates discretized by mesh 4×4;
(a) thin plate (t/L = 0.005); (b) thick plate (t/L = 0.1).

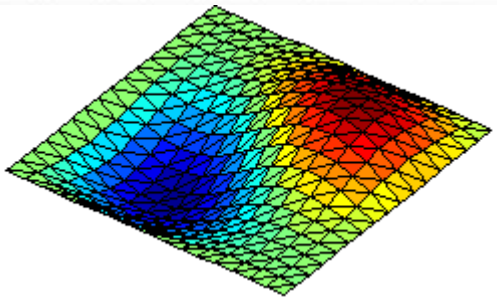
Phân tích mode tấm vuông



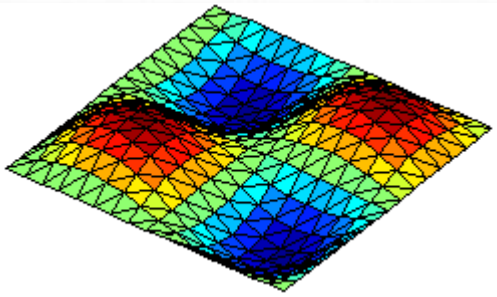
Mode 1



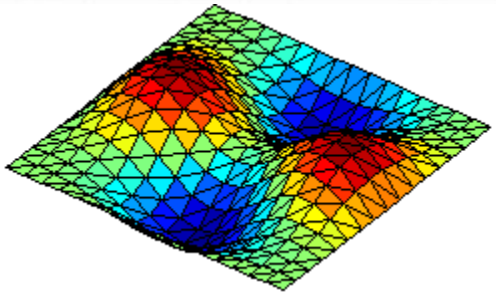
Mode 2



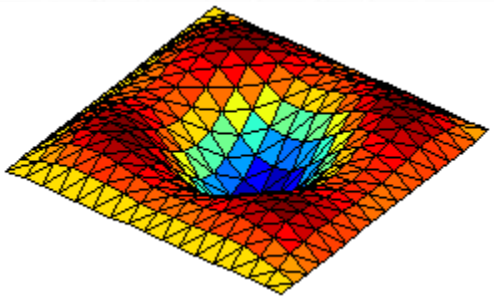
Mode 3



Mode 4

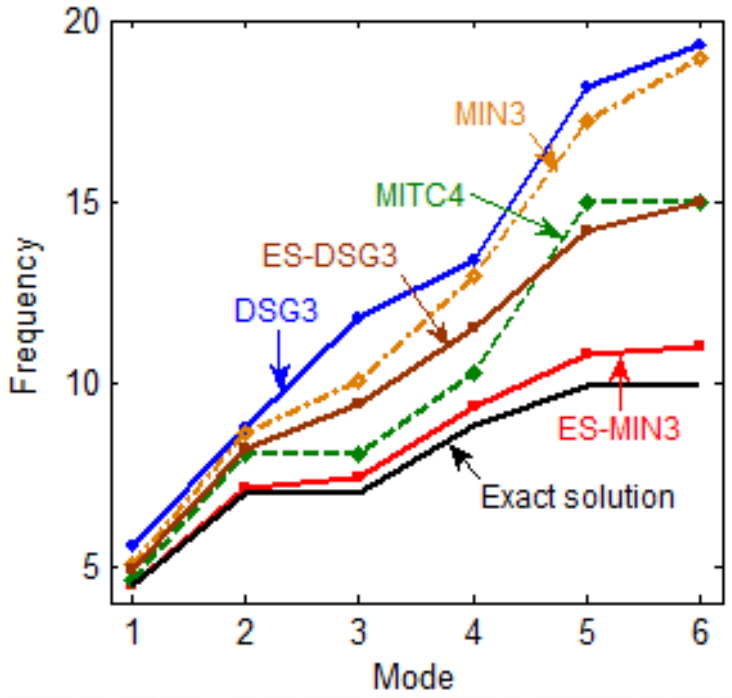


Mode 5

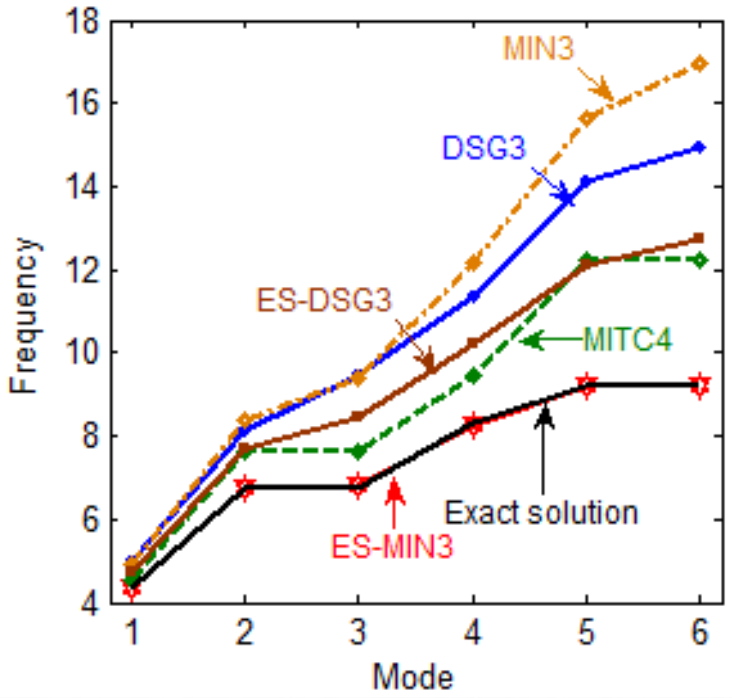


Mode 6

Phân tích mode tấm vuông



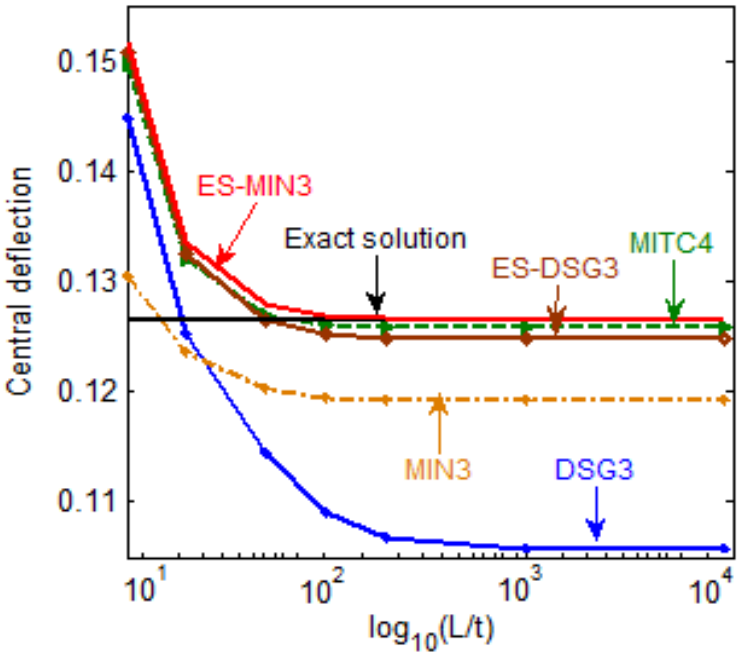
(a)



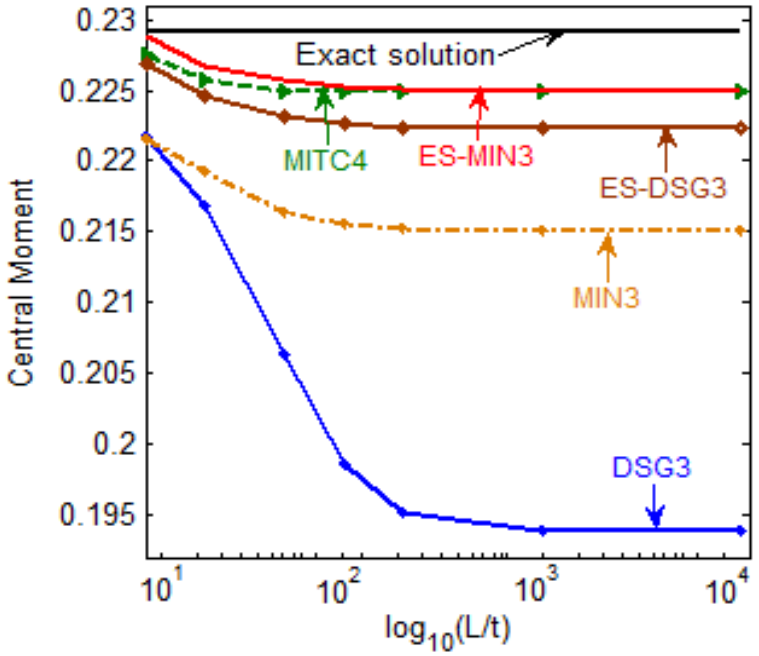
(b)

Six lowest frequencies of SSSS square plates discretized by mesh 4×4;
(a) thin plate (t/L = 0.005); (b) thick plate (t/L = 0.1).

Phân tích tĩnh tấm vuông



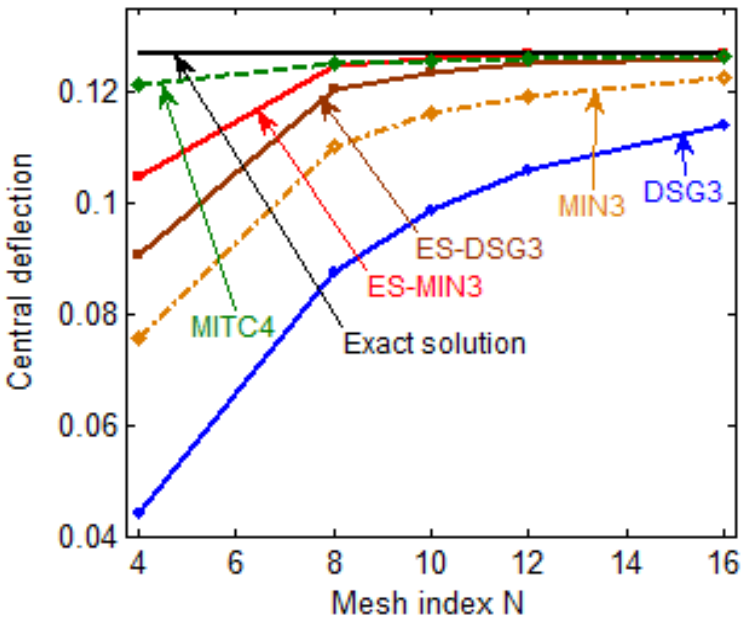
(a)



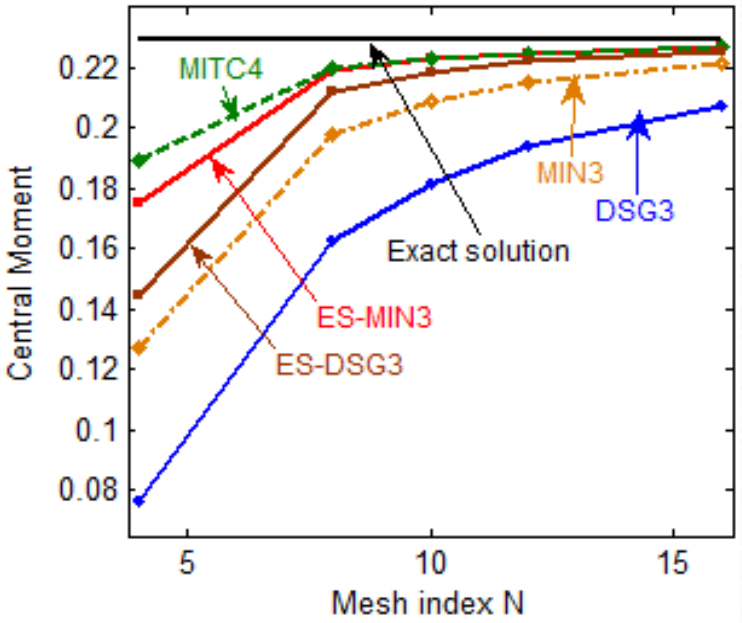
(b)

Results of various methods in the shear locking test of the clamped square plate subjected to uniform load with mesh 12x12; (a) central deflection; (b) central moment.

Phân tích tĩnh tấm vuông



(a)

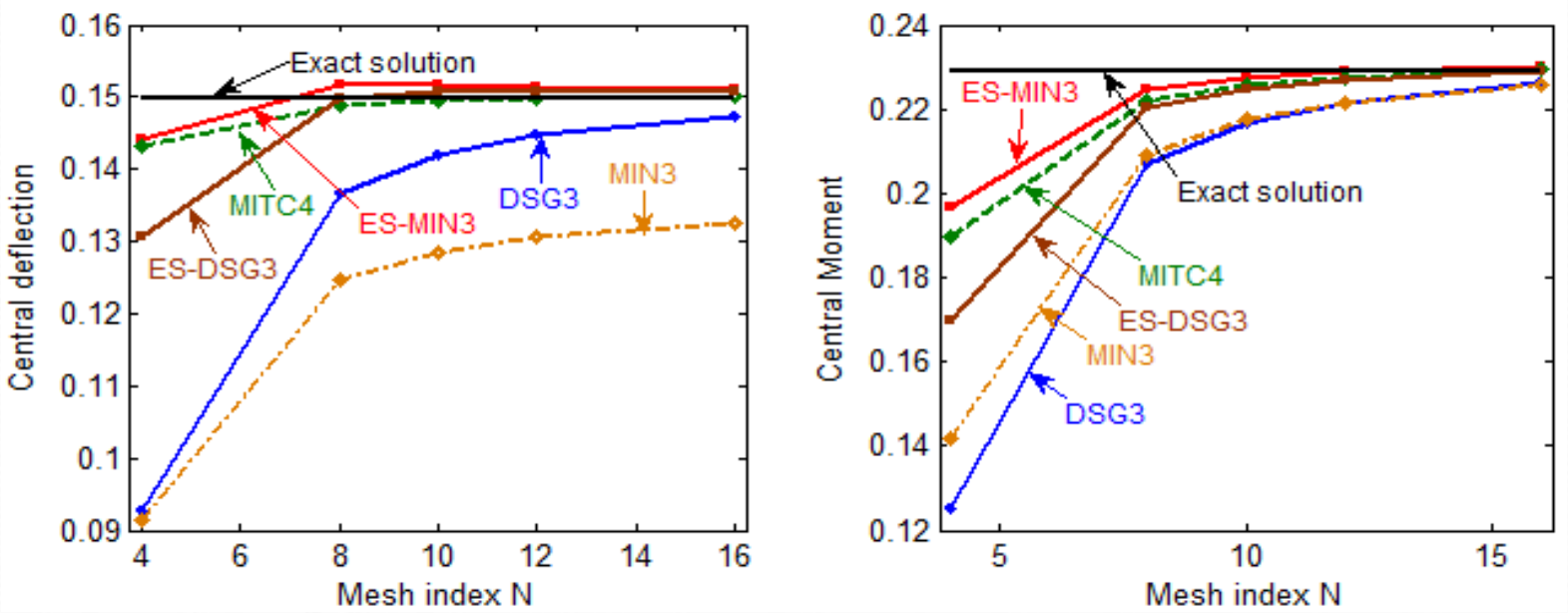


(b)

Convergence of results of clamped square plate with t/L=0.001;

(a) central deflection; (b) central moment.

Phân tích tĩnh tấm vuông

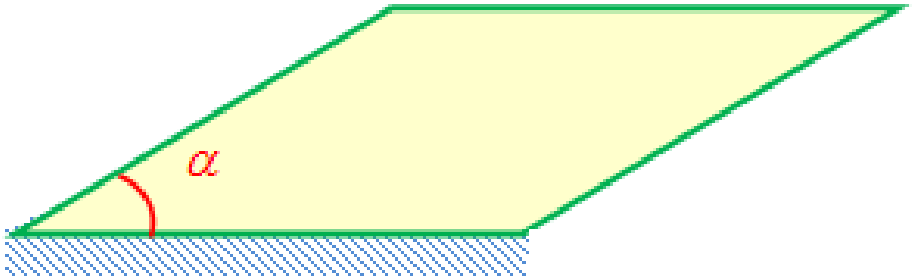


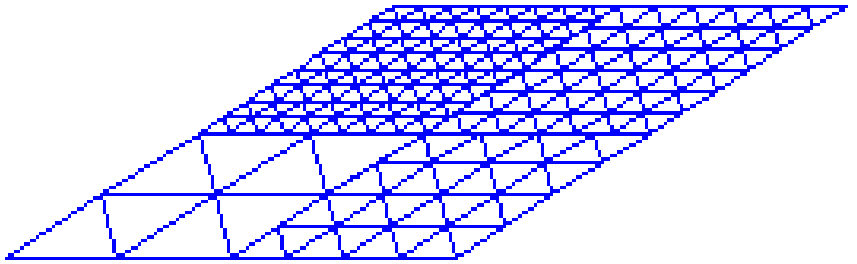
(a) (b)

Convergence of results of clamped square plate with t/L=0.1;

(a) central deflection; (b) central moment.

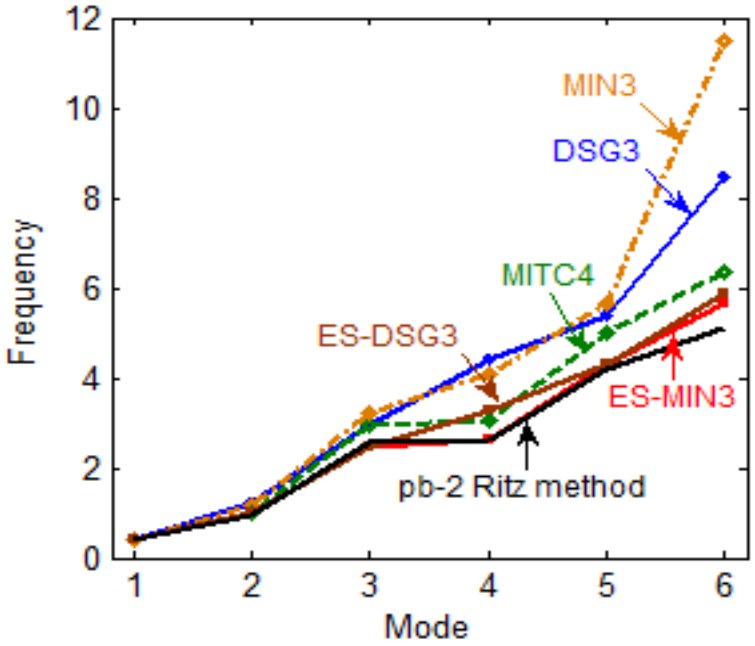
Tấm hình thoi



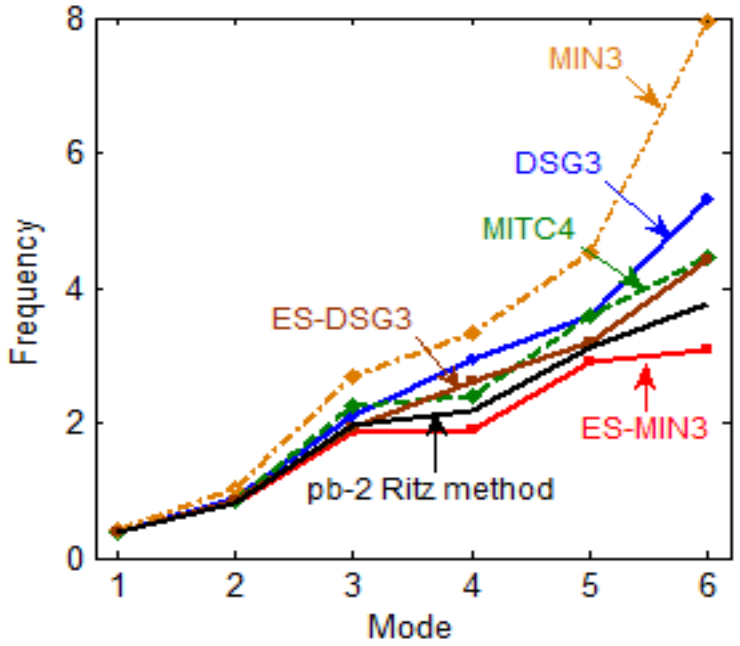


- ✓ Phân tích dynamic (mode)
- ✓ Phân tích static

Phân tích mode tấm hình thoi



(a)

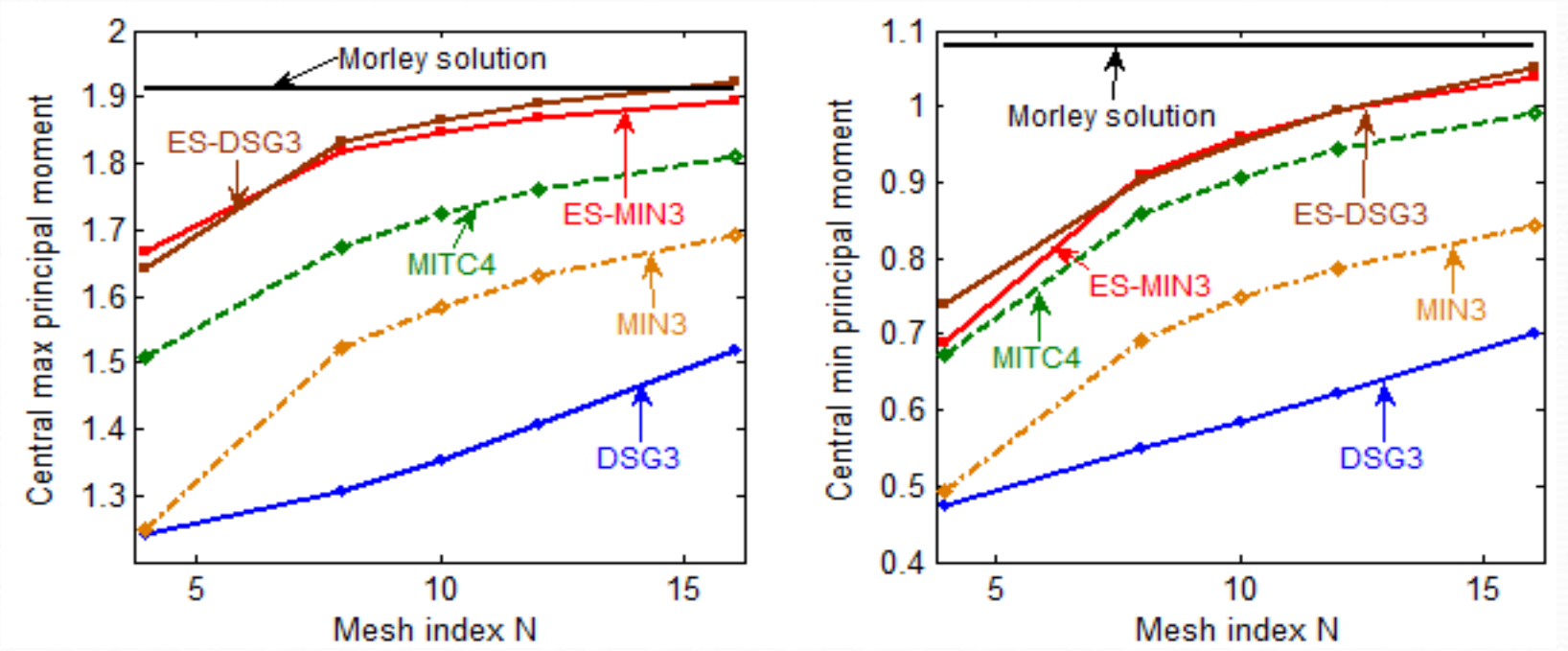


(b)

Six lowest frequencies of CFFF rhombic square plates discretized by mesh 4×4;

(a) thin plate (t/L = 0.001); (b) thick plate (t/L = 0.2).

Phân tích tĩnh tấm Morley

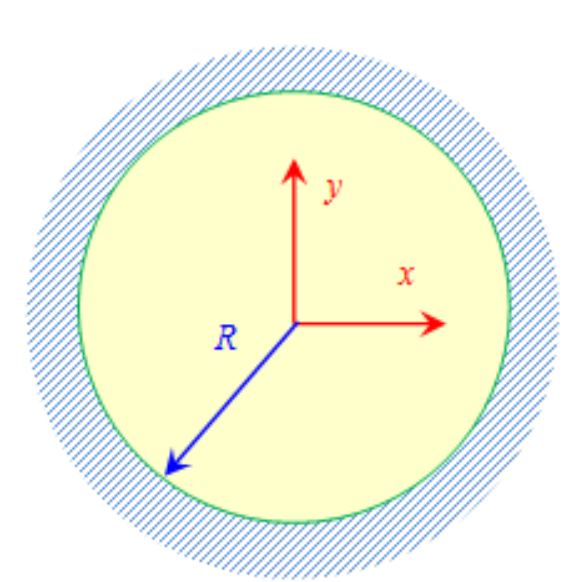


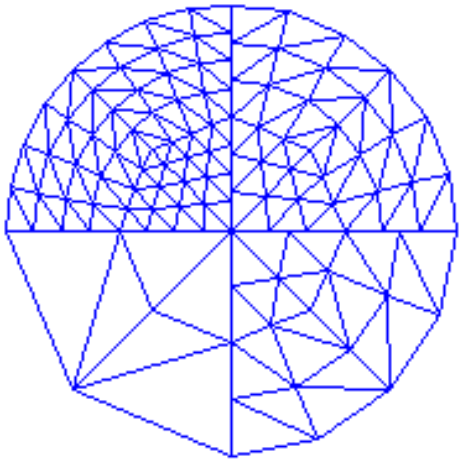
(a) (b)

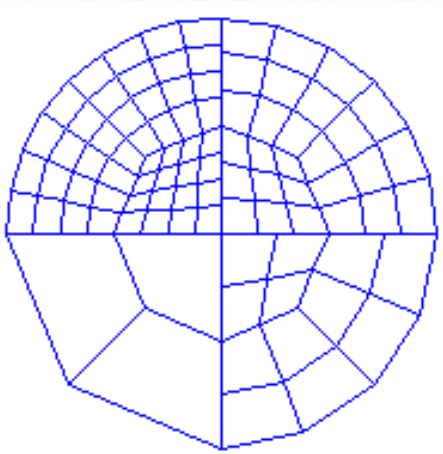
Convergence of results of at the central point of Morley plate (with L/t =1000);

(a) max principal moment; (b) min principal moment.

Tấm hình tròn

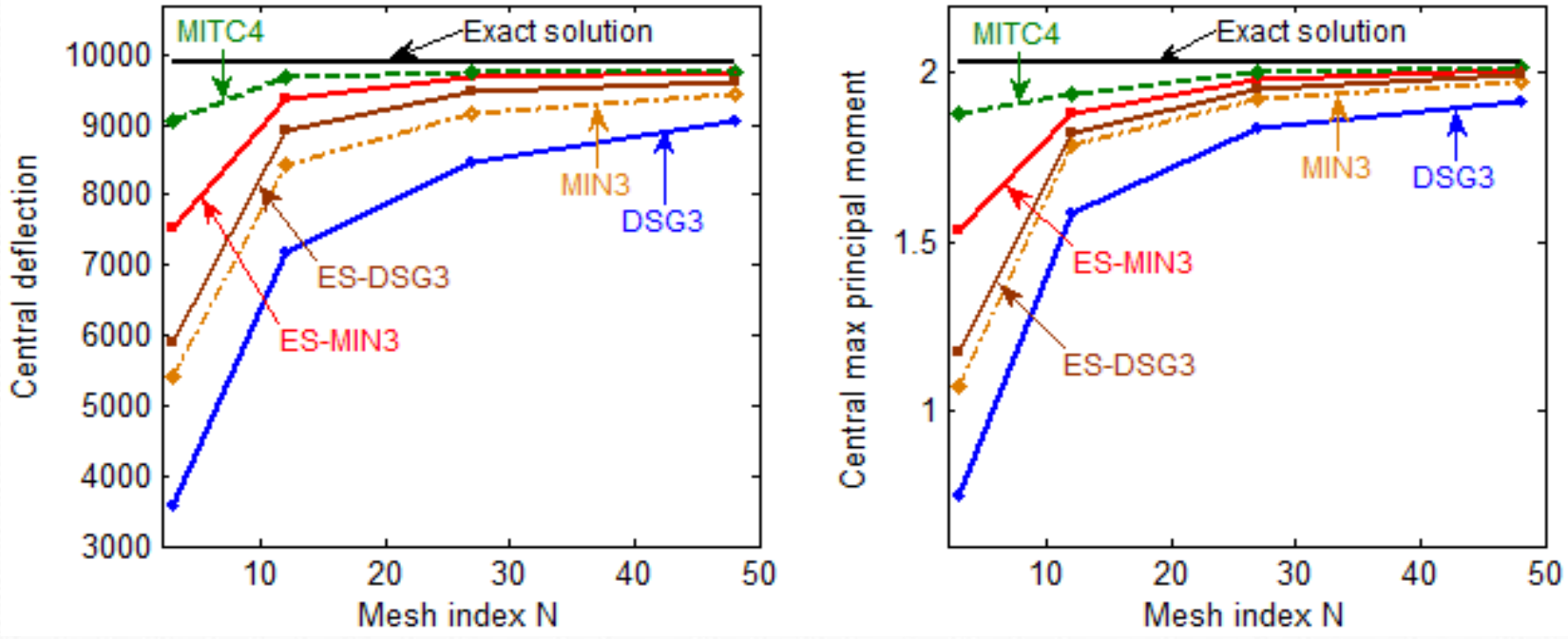






- ✓ Phân tích dynamic (mode)
- ✓ Phân tích static

Phân tích tĩnh tấm hình tròn

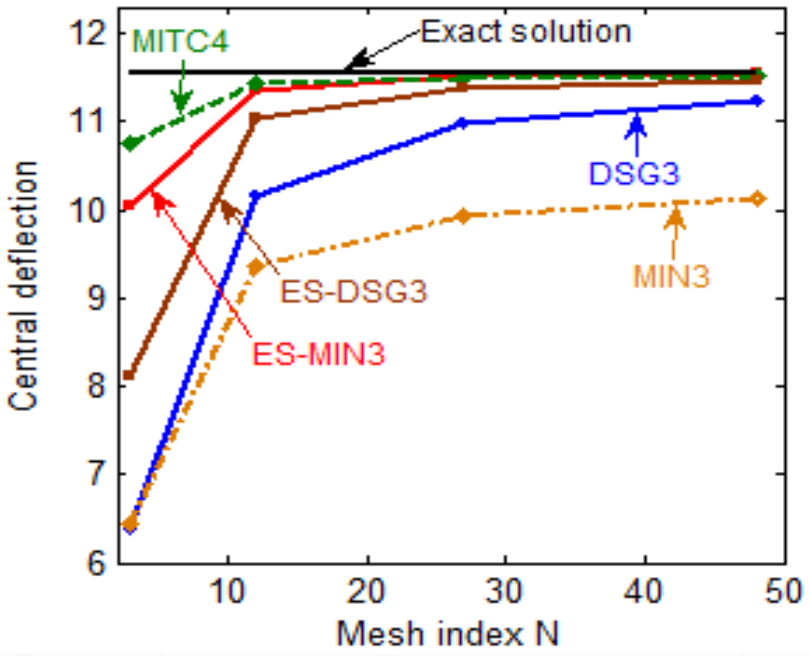


(a) (b)

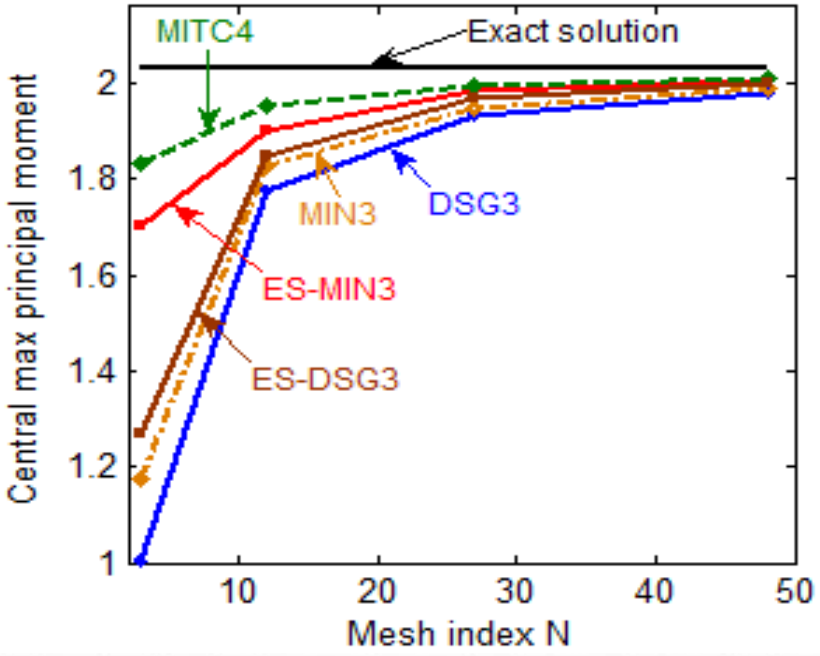
Convergence of results of clamped circular plate (t/R = 0.02) subjected to uniform load;

(a) central deflection; (b) central moment.

Phân tích tĩnh tấm hình tròn



(a)

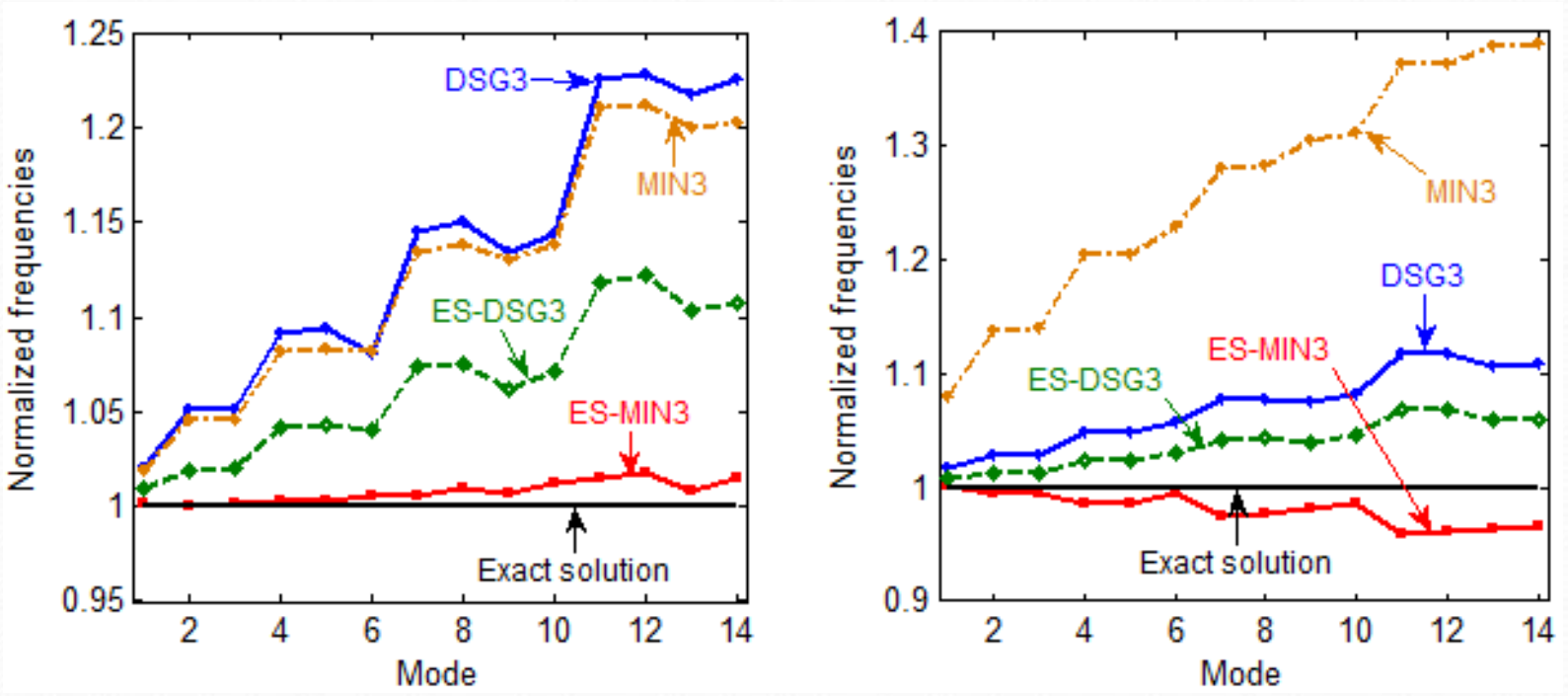


(b)

Convergence of results of clamped circular plate (t/R = 0.2) subjected to uniform load;

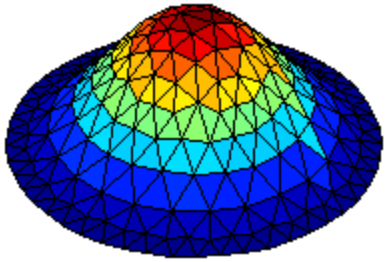
(a) central deflection; (b) central moment.

Phân tích mode tấm hình tròn

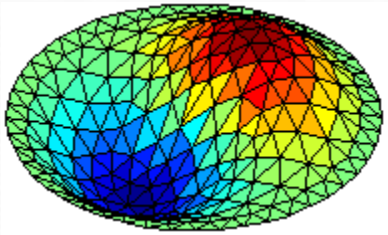


Comparison of fourteen normalized lowest frequencies (ω^h/ω_{exact}) of a clamped circular plate. (a) thin plate ($t/(2*R)$=0.01); (b) thick plate ($t/(2*R)$=0.1).

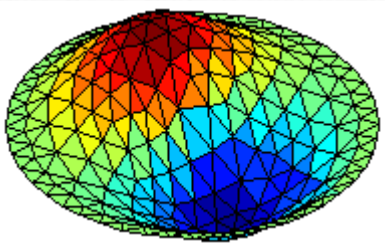
Phân tích mode tấm hình tròn



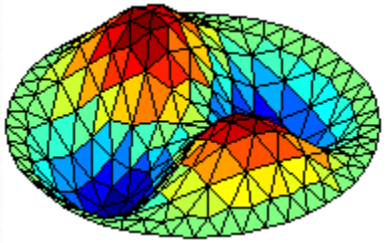
Mode 1



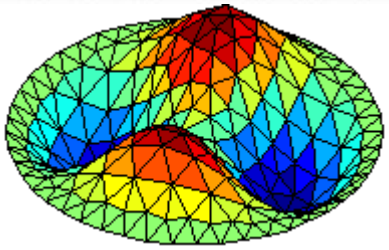
Mode 2



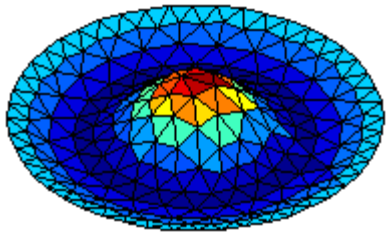
Mode 3



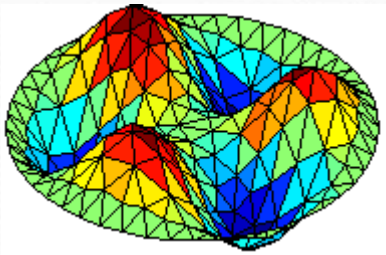
Mode 4



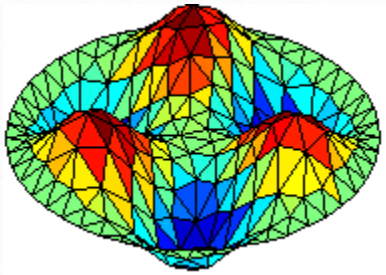
Mode 5



Mode 6

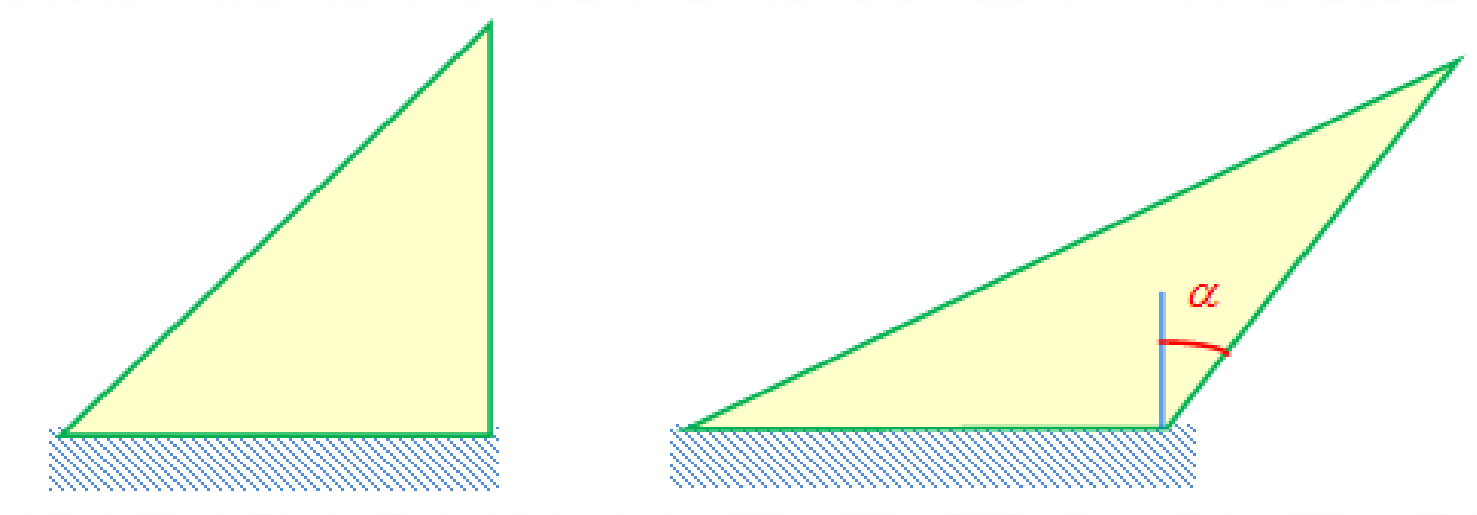


Mode 7



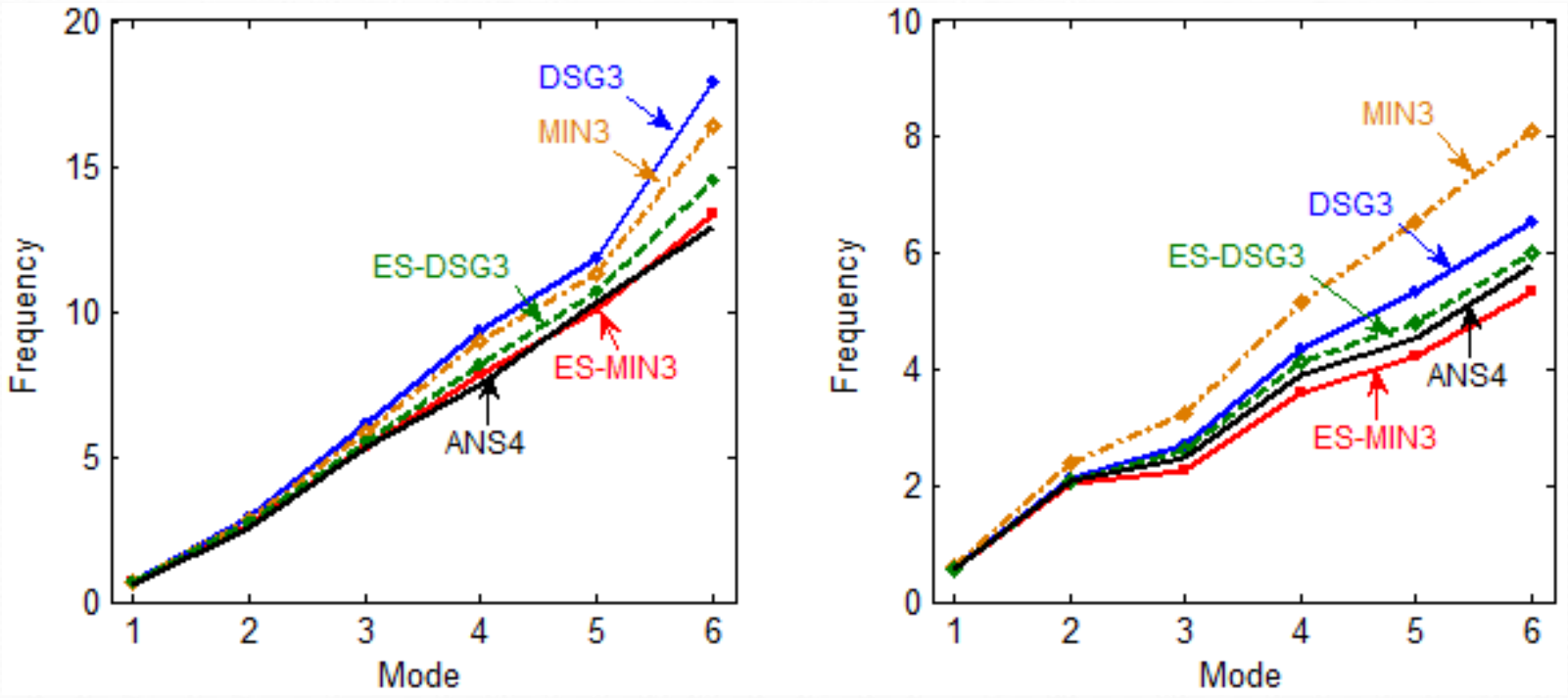
Mode 8

Tấm tam giác



✓ Phân tích dynamic (mode)

Phân tích mode tấm tam giác

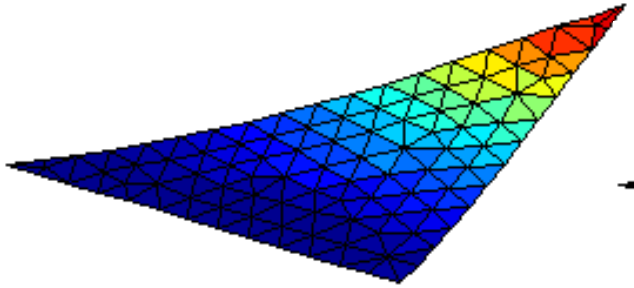


(a) (b)

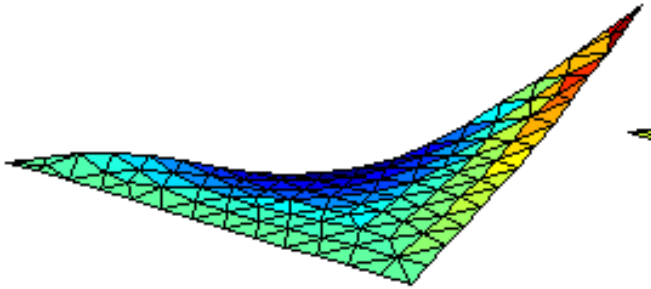
Comparison of six lowest frequencies of a thin triangular plate with the skew angle

(a) thin plate (t/L = 0.001); (b) thick plate (t/L = 0.2).

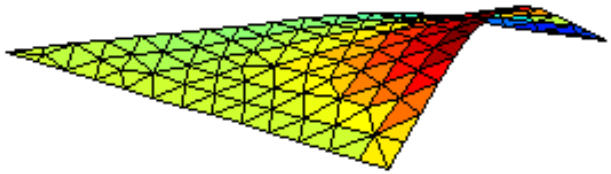
Phân tích mode tấm tam giác



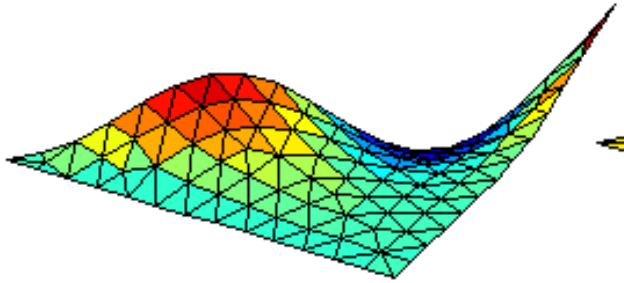
Mode 1



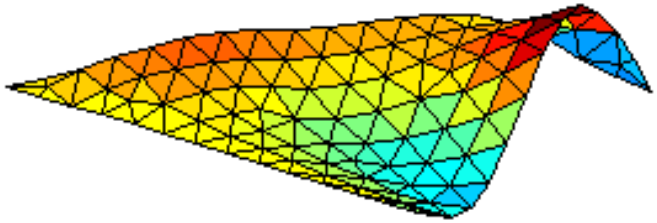
Mode 2



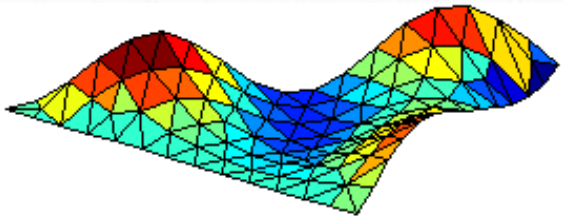
Mode 3



Mode 4



Mode 5



Mode 6

THANKS YOUR ATTENTION

